Inorganic CUME

April-1-2011

Graphene (a single graphite-sheet, also called "carbon in two-dimension" or "thinnest material possible") becomes a very hot topic in today's materials chemistry. Now your boss gives you a project to work on graphene.

1. What would be the color for a large single-sheet graphene? Why so? (10')

2. Sketch the pure graphene and graphene oxide for showing the structural difference between the two. (10')

3. Design a chemical process for converting the graphene oxide to a graphene chalcogenide. (20')

4. Design a chemical reaction to show how you could make a stable organometallic compound out of the graphene. (20')

5. Theoretically, could you make a graphene-wrapped C_{60}? Why or why not? (20')

6. Theoretically, could you make a graphene-$(BN)_n$ composite? Why or why not? (20')